

# Sublinear Time Spectral Density Estimation

---

Christopher Musco

New York University, Tandon School of Engineering

## COLLABORATORS



Vladimir Braverman  
(JHU)



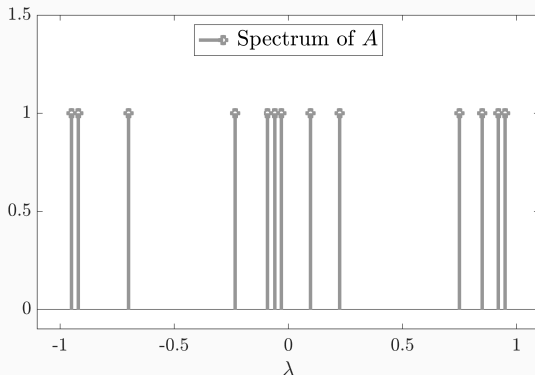
Aditya Krishnan  
(JHU)

Paper (work in progress) available at:  
<https://www.chrismusco.com/KPM-Chebyshev.pdf>.

# SPECTRAL DENSITY ESTIMATION

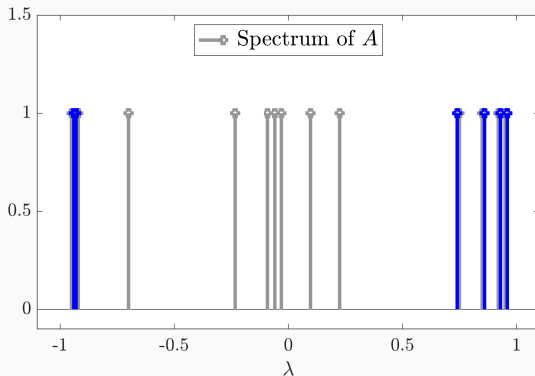
## Basic problem in linear algebra:

- Given a diagonalizable  $n \times n$  matrix  $\mathbf{A}$  with real eigenvalues  $\lambda_1, \dots, \lambda_n$ .
- Goal is to approximate this spectrum in  $\ll O(n^3)$  time.



# SPECTRAL DENSITY ESTIMATION

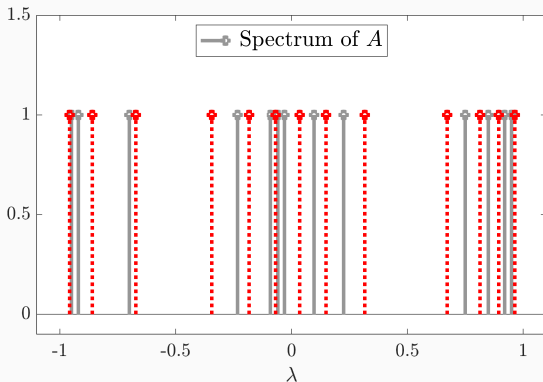
**Possible approach:** Compute a few outlying eigenpairs of  $A$  using an  $\sim O(n^2)$  time iterative method, like Lanczos iteration.



**Limitation:** Misses information about the interior of the spectrum.

# SPECTRAL DENSITY ESTIMATION

We want to capture information about the whole spectrum.

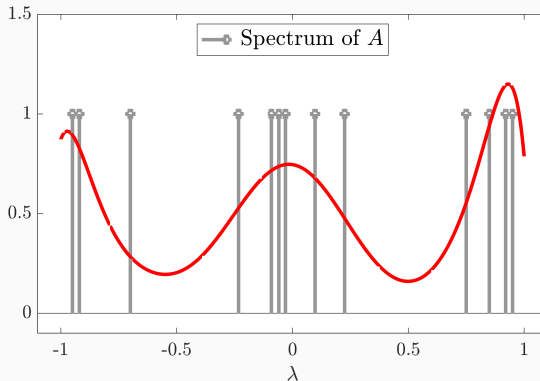


**Possible goal:** Let  $\boldsymbol{\lambda} = [\lambda_1, \dots, \lambda_n]$ . Return approximate eigenvalues  $\tilde{\boldsymbol{\lambda}} = [\tilde{\lambda}_1, \dots, \tilde{\lambda}_n]$  such that:

$$\|\boldsymbol{\lambda} - \tilde{\boldsymbol{\lambda}}\| < \epsilon.$$

# SPECTRAL DENSITY ESTIMATION

Would also be happy with a “smooth” approximation to the eigenvalue distribution



Easily discretized if approximate eigenvalues are desired.

## EXAMPLE APPLICATIONS

- Computational physics and chemistry.
- Subroutine used to initialize parallel eigensolvers, like the FEAST eigensolver [Polizzi, 2009].
- Analyzing the spectra of weight matrices and Hessian matrices in deep learning. Understanding generalization, improving convergence, etc. [Mahoney et. al. 2019+].
- Analyzing graph structure in network science. E.g. the adjacency matrix of a social network graph. [Dong, Benson, Bindel, 2019].

## EXISTING METHODS

There has been a lot of work on this problem, and many methods proposed to solve it.

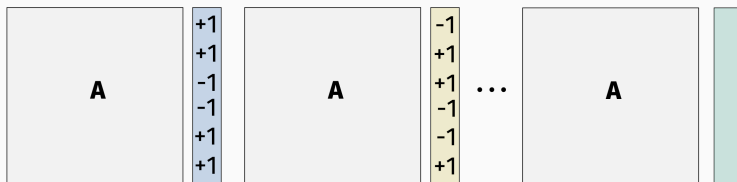
- Kernel Polynomial Method
- Lanczos Spectroscopic Method
- Delta-Gauss-Legendre quadrature
- Lanczos Method for CDF

See [\[Lin, Saad, Yang, 2014\]](#) for a good overview.

**Bottom line:** Many of these methods are very similar, or exactly equivalent. Currently difficult to compare, beyond experimental results involving specific applications.

## EXISTING METHODS

All based on repeated matrix-vector multiplications with  $A$ , usually with random vectors.



Run in roughly  $T \cdot n^2$  time, which is linear time for a dense  $n \times n$  matrix. But it's not clear how large  $T$  needs to be for a desired level of approximation.

One recent method avoids the  $O(n^2)$  cost for certain classes of matrices, running in sublinear time.

- [Cohen-Steiner, Kong, Sohler, Valiant, 2018] gives a method for normalized graph adjacency and Laplacian matrices assuming sample access to the graph.
- In  $2^{O(1/\epsilon)}$  time returns approximate eigenvector list with  $\frac{1}{n}\|\boldsymbol{\lambda} - \tilde{\boldsymbol{\lambda}}\|_1 < \epsilon$ . **Not practical, but very interesting!** There is no dependence on  $n$ .

How do we compare this result to existing algorithms?  
Develop entirely new techniques?

## GOAL FOR TODAY

1. Define a natural objective for the spectral density estimation problem based on approximation in the Wasserstein-1 metric (used several times before).
2. Show that a popular variant of the Kernel Polynomial Method (KPM) based on the Jackson kernel achieves an  $\epsilon$  approximation in  $\sim O(n^2/\epsilon)$  time – i.e.,  $O(1/\epsilon)$  matvecs.
3. Describe a practical, randomized sublinear time algorithm for normalized graph adjacency and Laplacian matrices that achieves an  $\epsilon$  approximation in  $O(n/\epsilon^c)$  time.

Hoping to initiate the study of a greater variety of algorithmic techniques for spectral density estimation.

## HOW TO MEASURE SOLUTION ACCURACY

View spectrum as a probability density. If  $\mathbf{A}$  has eigenvalues  $\lambda_1, \dots, \lambda_n$ ,

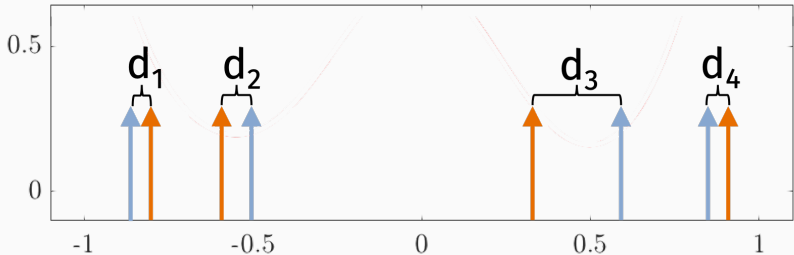
Spectral density: 
$$s(x) = \frac{1}{n} \sum_{i=1}^n \delta(x - \lambda_i).$$

**Goal:** Find density  $q$  which is close to  $s$  in some metric.

**Natural choice:** Wasserstein-1 distance  $W_1(s, q)$ . Aka “earth mover’s distance”.

# WASSERSTEIN DISTANCE

Compute cost of optimal transport plan for moving one distribution to another. E.g. for two point-mass distributions:

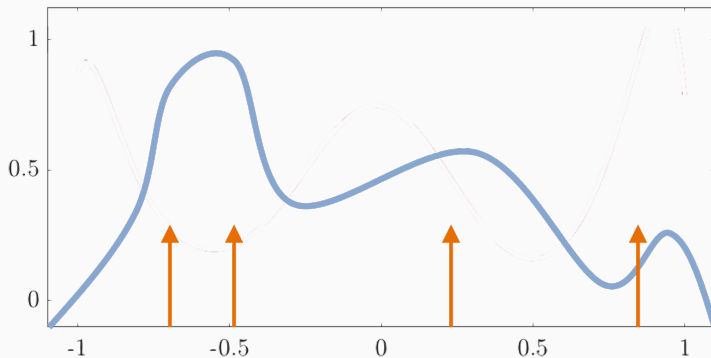


$$W_1(s, q) = \frac{1}{4}(d_1 + d_2 + d_3 + d_4).$$

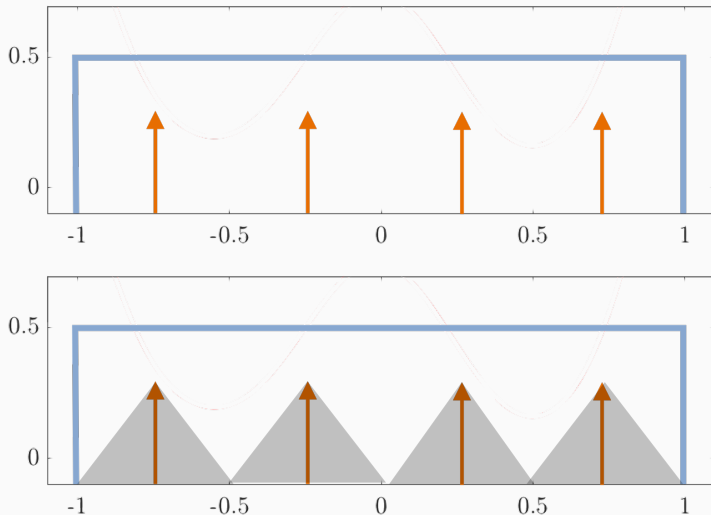
For spectral densities  $s, q$  with eigenvalues  $\boldsymbol{\lambda}$  and  $\tilde{\boldsymbol{\lambda}}$ ,  
 $W_1(s, q) = \frac{1}{n} \|\boldsymbol{\lambda} - \tilde{\boldsymbol{\lambda}}\|_1$ . Equivalent to Cohen-Steiner et al. metric.

# WASSERSTEIN DISTANCE

Also easily used to compare continuous and point mass distributions.



# WASSERSTEIN DISTANCE



$$W_1(s, q) = 1/6 < 1/2.$$

Assume distribution are supported on  $[-1, 1]$  and let  $f$  be a 1-Lipschitz function from  $[-1, 1] \rightarrow \mathbb{R}$ .

I.e.  $|f(x) - f(y)| < |x - y|$  for all  $x, y$ .

**Dual characterization:**

$$W_1(s, q) = \max_{1\text{-Lipschitz } f} \langle f, s - q \rangle$$

where  $\langle a, b \rangle = \int_{-1}^1 a(x)b(x)dx$ .

Existing SDE algorithms are based on the idea that we can efficiently approximate moments of the spectral density  $s$  by relating them to the trace of matrix polynomials involving  $A$ .

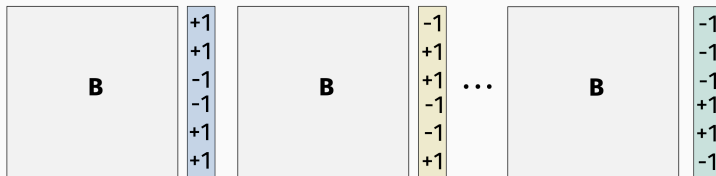
Let  $p$  be a low-degree polynomial. We have:

$$\langle s, p \rangle = \int_{-1}^1 s(x)p(x)dx = \frac{1}{n} \sum_{i=1}^n p(\lambda_i) = \frac{1}{n} \text{tr}(p(\mathbf{A})),$$

where  $p(\mathbf{A}) = c_0 \mathbf{I} + c_1 \mathbf{A} + \dots, + c_q \mathbf{A}^q$ .

Hutchinson 1991, Girard 1987:

- Draw  $\mathbf{x}_1, \dots, \mathbf{x}_m \in \mathbb{R}^n$  i.i.d. with random  $\{+1, -1\}$  entries.
- Return  $\tilde{T} = \frac{1}{m} \sum_{i=1}^m \mathbf{x}_i^T \mathbf{B} \mathbf{x}_i$  as approximation to  $\text{tr}(\mathbf{B})$ .



Requires  $m$  matvecs with  $\mathbf{B}$ . When  $\mathbf{B} = p(\mathbf{A})$  for degree  $q$  polynomial, each matvec with  $\mathbf{B}$  requires  $q$  matvecs with  $\mathbf{A}$ .

Let  $\tilde{T}$  be the trace estimate returned by Hutchinson's method.

**Claim (Rudelson, Vershynin, 2013, Roosta, Ascher 2015)**

*If  $m = O\left(\frac{\log(1/\delta)}{\epsilon^2}\right)$ , then with probability  $(1 - \delta)$ ,*

$$\left| \tilde{T} - \text{tr}(\mathbf{B}) \right| \leq \epsilon \|\mathbf{B}\|_F.$$

## MOMENT MATCHING

So, roughly, we can compute  $\langle s, p \rangle = \frac{1}{n} \text{tr}(p(\mathbf{A}))$  with  $O(q)$  matvecs if  $p$  is a degree  $q$  polynomial.

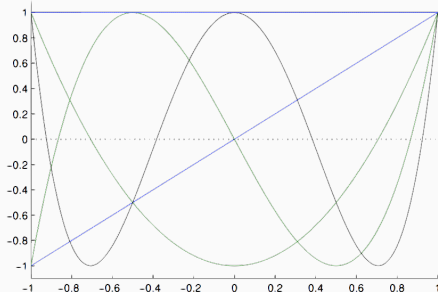
**Classical statistical approach:** Find any distribution that matches  $s$  on first  $N$  standard moments. So compute trace for  $p_1(x) = x, p_2(x) = x^2, \dots, p_N(x) = x^N$ .

**[Kong, Valiant 2017]:** If distributions  $s$  and  $q$  match exactly on their first  $N = O(1/\epsilon)$  moments, then  $W_1(s, q) \leq \epsilon$ .

**Main issue:** This is a “poorly conditioned” statement. Small differences in the moments can quickly weaken the bound.

## SERIES EXPANSION

Let  $T_0, \dots, T_N$  be the (normalized) Chebyshev polynomials.



Recall that these polynomials are orthogonal under the weight:

$$w(x) = \frac{1}{\sqrt{1-x^2}}.$$

I.e.  $\langle T_k, w \cdot T_k \rangle = 1$ , and for  $k \neq j$ ,  $\langle T_k, w \cdot T_j \rangle = 0$ .

Better approach [Weiß, Wellein, Alvermann, Fehske 2006].

Let  $\tau_k = \langle s, T_k \rangle = \frac{1}{n} \text{tr}(T_k(\mathbf{A}))$ . The Chebyshev series for  $s/w$  is:

$$\frac{s}{w} = \sum_{k=1}^{\infty} \tau_k T_k.$$

Approximate via truncation as:

$$\frac{s}{w} \approx \sum_{k=1}^N \tau_k T_k = g_N,$$

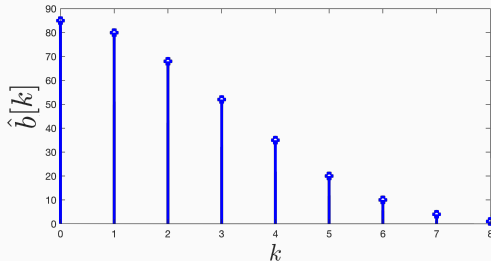
and return the approximation:

$$s_N = w \cdot g_N.$$

**Problem:** Truncated Chebyshev series is unlikely to be positive, even if  $s$  and  $s/w$  are. We want it to return a density.

Instead use a damped Chebyshev series due to [\[Jackson, 1912\]](#).

$$\frac{s}{w} \approx \sum_{k=1}^N \frac{\hat{\beta}_k}{\hat{\beta}_0} \tau_k T_k,$$



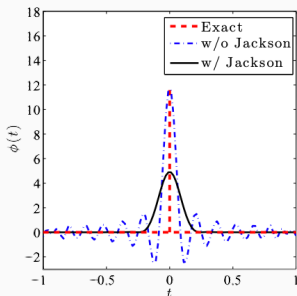
$\hat{b} = \text{conv}(\text{conv}(g, g), \text{conv}(g, g))$  where  $g$  is an all ones vector.

**Properties of Jackson kernel.** For any function  $f$  with

$\tau_k = \langle f, w \cdot T_k \rangle$ , let  $f_N = \sum_{k=1}^N \frac{\hat{\beta}_k}{\hat{\beta}_0} \tau_k T_k$ . We have:

1.  $f_N(x) \geq 0$  for all  $x \in [-1, 1]$  if  $f$  is non-negative.
2.  $\|f - f_N\|_\infty \leq O(\lambda/N)$  if  $f$  is  $\lambda$ -Lipschitz (Jackson's theorem).

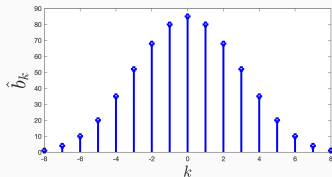
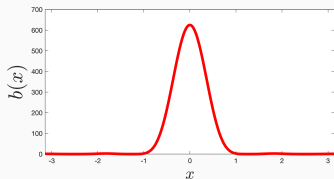
First property is immediately useful, since we apply this series to the function  $s/w$  which is non-negative.



From [Lin, Saad, Yang, 2014].

## Where does Jackson's kernel come from?

1. Extend  $f$  to an even, periodic function  $g$  via transformation  $g(\theta) = r(\cos\theta)$  for  $\theta \in [-\pi, \pi]$ . Fourier series coefficient of  $g$  are equal to the Chebyshev series coefficients of  $f$ .
2. Convolve  $g$  with “bump function”  $b = \left( \frac{\sin(mx/2)}{\sin(x/2)} \right)^4$ , which has bounded Fourier support.
3. Equivalent to multiplying series coefficients by those of  $b$ .



Bump function  $b(x)$  for  $m = 5$  and its Fourier series coefficients.

Why does this approach give a good approximation?  $s/w$  isn't continuous, let alone Lipschitz.

Let  $g = s/w$ , and let  $f_N, g_N$  be Jackson series approximations to  $f, g$ . Let  $s_N = wg_N$  (what the KPM returns).

**Key observation:** For any  $f$ ,

$$\langle f, s - s_N \rangle = \langle f - f_N, s \rangle.$$

Proof. Let  $\mathcal{T}$  be quasimatrix with  $\mathcal{T}_{k,x} = T_k(x)$  and let  $\mathcal{W}$  be a diagonal operator with  $\mathcal{W}_{x,x} = \frac{1}{\sqrt{1-x^2}}$ . Let  $\beta$  be a diagonal operator with  $\beta_{i,i} = \hat{b}_i$ ,  $\beta_{i,i} = 0$  for  $i > N$ . We have:

$$f_N = \mathcal{T}^T \beta \mathcal{T} \mathcal{W} f \qquad g_N = \mathcal{T}^T \beta \mathcal{T} \mathcal{W} g.$$

$$\langle f, s_N \rangle = \langle f, \mathcal{W} \mathcal{T}^T \beta \mathcal{T} \mathcal{W} g \rangle = \langle \mathcal{W} \mathcal{T}^T \beta \mathcal{T} f, \mathcal{W} g \rangle = \langle f_N, s \rangle$$

An approximation guarantee immediately follows for  $s_N$  returned by the kernel polynomial method:

$$W_1(s, s_N) = \max_{1\text{-Lipschitz } f} \langle f, s - s_N \rangle = \max_{1\text{-Lipschitz } f} \langle f - f_N, s \rangle \leq \epsilon$$

as long as we choose  $N = O(1/\epsilon)$ .

All we need to do is implement a routine for approximating  $\frac{1}{n} \text{tr}(T_k(A))$  for  $k = 1, \dots, N$  up to additive error  $\epsilon^2$ .

**Claim:** Suffices to run Hutchinson's estimator with  $\ell = \max(1, \epsilon^{-4}/n)$  iterations.<sup>1</sup>

In total, Jackson Kernel KPM provides an  $\epsilon$ -approximate SDE with  $O(\ell \cdot N) = O(\ell/\epsilon^2)$  matvecs with  $A$ .

---

<sup>1</sup>Result improves with  $n$  because Hutchinson's error scales as  $\|A\|_F \leq \sqrt{n}$ .

## A SUBLINEAR TIME RESULT FOR GRAPHS

Main subroutine of Hutchinson's method:

Compute  $\mathbf{v}_0 = T_0(\mathbf{A})\mathbf{g}$ ,  $\mathbf{v}_1 = T_1(\mathbf{A})\mathbf{g}$ ,  $\dots$ ,  $\mathbf{v}_N = T_N(\mathbf{A})\mathbf{g}$  for random vector  $\mathbf{g}$ . Easiest way if via three-term recurrence:

- $\mathbf{v}_0 = \mathbf{g}$
- $\mathbf{v}_1 = \mathbf{A}\mathbf{g}$ .
- For  $k \geq 2$ ,
  - $\mathbf{v}_k = 2\mathbf{A}\mathbf{v}_{k-1} - \mathbf{v}_{k-2}$ .

This recurrence is well-known to have good stability properties [Clenshaw, 1955].

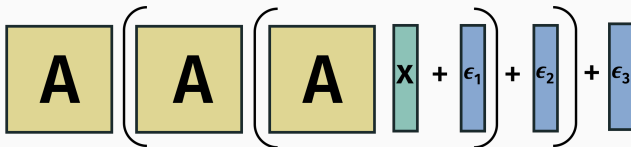
**Idea:** Replace every matvec with  $\mathbf{A}$  by an approximate matrix vector multiply. **The approximation can be quite coarse!**

## ERROR ACCUMULATION

Suppose we have a routine  $AMV(\mathbf{A}, \mathbf{x})$  which for any vector  $\mathbf{x}$  ensures that:

$$\|AMV(\mathbf{A}, \mathbf{x})\|_2 \leq \epsilon_{MV} \|\mathbf{x}\|_2.$$

If we use this routine to compute  $\mathbf{A}^q \mathbf{x}$ , error builds linearly.



Obtain approximation  $\mathbf{v}$  with  $\|\mathbf{A}^q \mathbf{x} - \mathbf{v}\|_2 \leq \epsilon_{MV} q \|\mathbf{x}\|_2$

## ERROR ACCUMULATION

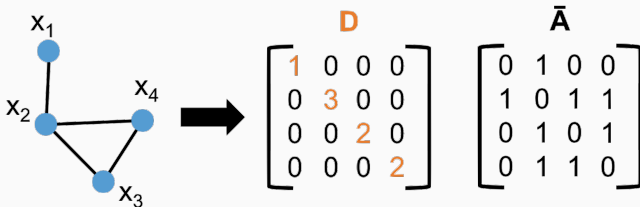
Suppose we use the routine to compute Chebyshev polynomials:

- $\mathbf{v}_0 = \mathbf{g}$
- $\mathbf{v}_1 = \text{AMV}(\mathbf{A}, \mathbf{g})$ .
- For  $k \geq 2$ ,
  - $\mathbf{v}_k = 2\text{AMV}(\mathbf{A}, \mathbf{v}_{k-1}) - \mathbf{v}_{k-2}$ .

Then using an argument of Clenshaw's based on Chebyshev polynomials of the second kind, it is possible to show that error builds quadratically.

Obtain approximation  $\mathbf{v}$  with  $\|T_q(\mathbf{A})\mathbf{x} - \mathbf{v}\|_2 \leq \epsilon_{\text{MV}} O(q^2) \|\mathbf{x}\|_2$ . In our setting,  $q = O(1/\epsilon)$ , so need to set  $\epsilon_{\text{MV}} = O(\epsilon^3)$ .

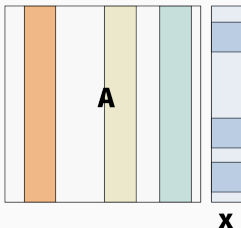
## APPROXIMATE MATVECS FOR ADJACENCY MATRICES



Normalized adjacency matrix equals  $D^{-1/2}\bar{A}D^{-1/2}$ . Eigenvalues bounded between  $[-1, 1]$ . We actually consider  $A = \bar{A}D^{-1}$  which has the same spectral density as the normalized adjacency.

## APPROXIMATE MATVECS FOR ADJACENCY MATRICES

Approximate  $\mathbf{Ax}$  by random subsampling each column with probability  $p_i$ :



$$\mathbf{Ax} \approx \sum_{j=1}^T \frac{1}{p_{i_j}} \mathbf{A}^{i_j} x_{i_j}$$

**Drineas, Kannan, Mahoney, 2006.** If we set  $p_i = \frac{1}{\Delta^2} \frac{\|\mathbf{A}^i\|_2}{\|\mathbf{A}\|_F^2}$  then with probability 9/10,

$$\|\text{AMV}(\mathbf{A}, \mathbf{x}) - \mathbf{Ax}\|_2 \leq \Delta \|\mathbf{A}\|_F \|\mathbf{x}\|_2.$$

Need to set  $\Delta = \epsilon_{\text{MV}} / \|\mathbf{A}\|_F$ .

**Key observation:** Norm of column  $i$  is  $\|\mathbf{A}^i\|_2^2 = \frac{1}{d_i}$ . So we sample denser columns with lower probability.

Expected number of non-zeros is columns sampled:

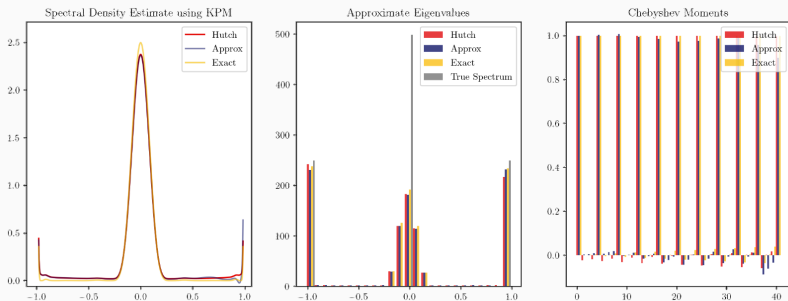
$$\sum_{i=1}^n p_i \frac{1}{\|\mathbf{A}^i\|_2^2} = \sum_{i=1}^n \frac{\|\mathbf{A}^i\|_2^2}{\epsilon_{MV}^2} \cdot \frac{1}{\|\mathbf{A}^i\|_2^2} = \frac{n}{\epsilon_{MV}^2}.$$

So we can compute  $\text{AMV}(\mathbf{A}, \mathbf{x})$  in just  $O(n/\epsilon_{MV}^2)$  time!

## FINAL RESULT

Combine approximate matrix-vector multiplications algorithm, Hutchinson's estimator, and KPM to obtain:

$O(n/\epsilon^c)$  time SDE method for normalized adjacency and Laplacian matrices. Promising initial results.



$W_1(s, q) < .05$  in 30% of the time of full KPM method.

## OPEN QUESTIONS

- Compare to  $2^{O(1/\epsilon)}$  time for Cohen-Steiner et al. method. Is  $O(1/\epsilon^c)$  time possible for normalized adjacency matrices?  $O(\sqrt{n}/\epsilon^c)$ ?
- Are there other classes of matrices where sublinear time results are possible?
- Do other common SDE algorithms achieve  $\epsilon$ -error in the Wasserstein-1 distance with a comparable number of matvecs? Is this even the right metric to care about?

THANKS! QUESTIONS?